Revenue Growth Management (RGM) Analytics

Power BI Semantic Model, Measures & Analytics Design Document

Waves 1–3 Complete

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Version: 3.0 (Final Release)

Power BI Analytics

Project Context & Objective

CONTEXT

This Power BI implementation is the **final consumption layer** of the RGM analytics pipeline. It transforms validated datasets from upstream SQL/R processing into interactive decision support tools.

SQL Data Warehouse
↓
R Processing (Logic)
↓
Power BI (Visual Layer)

PRIMARY OBJECTIVES

Single Source of Truth

Establish a unified, validated set of commercial KPIs to eliminate discrepancies in reporting across regions and channels.

Diagnostic Intelligence

Enable deep-dive diagnostics into pricing leakage, trade spend efficiency, and margin contribution drivers.

Trend Monitoring

Support weekly cadence reviews with rolling trend analysis to identify performance momentum and risks early.

Executive Readiness

Deliver polished, high-level performance views suitable for immediate consumption by senior commercial leadership.

Data Architecture: Fact and Dimensions

Fact Table: rgm_weekly_kpis

Grain: Weekly

Aggregated commercial performance data

DIMENSIONAL KEYS

- Product Family
- Operating Country
- Channel
- Wholesaler Tier
- Fiscal Week

CORE METRICS

- Revenue (Gross & Net)
- Trade Spend
- Volume (Shipments)
- Pricing
- Margin Proxy

Dimension Tables

LOADED DIMENSIONS

- dim_date_week
- dim_country
- dim_region
- dim_channel
- dim_wholesaler
- dim_product
- dim_product_family NEW

Data Validation & Quality

- All datasets validated for correct data types
- Headers cleaned and standardized
- Business-aligned column naming conventions applied
- No analytical transformations in Power Query (Pure ELT)

Product Family Dimension Design

1

Issue Identified

Ambiguous Grain: The source product dimension (dim_product) contained multiple rows per product family.

Modeling Risk: Variants and formats caused duplicate family entries, creating many-to-many relationship risks.

Product_ID	Family
P-201	Oral Nicotine
P-202	Oral Nicotine

2

Solution Implemented

DAX Calculated Table: Created a dedicated `dim_product_family` table directly in Power BI.

Transformation: Referenced base table, selected family column, and applied DISTINCT to remove duplicates.

```
Family = DISTINCT(
'dim_product'[Family]
)
```

3

Final Results

Unique Keys: Achieved one row per product family with a clean business key.

Optimized Model: Fact table connects only to this new dimension, aligning perfectly with commercial decision grain.

Family_Key	Family_Name
3	Heated Tobaccoo
2	Cigarettes
1	Oral Nicotine

Semantic Model Structure

Modeling Principles

Star Schema Design

Single fact table design with one-to-many relationships only.

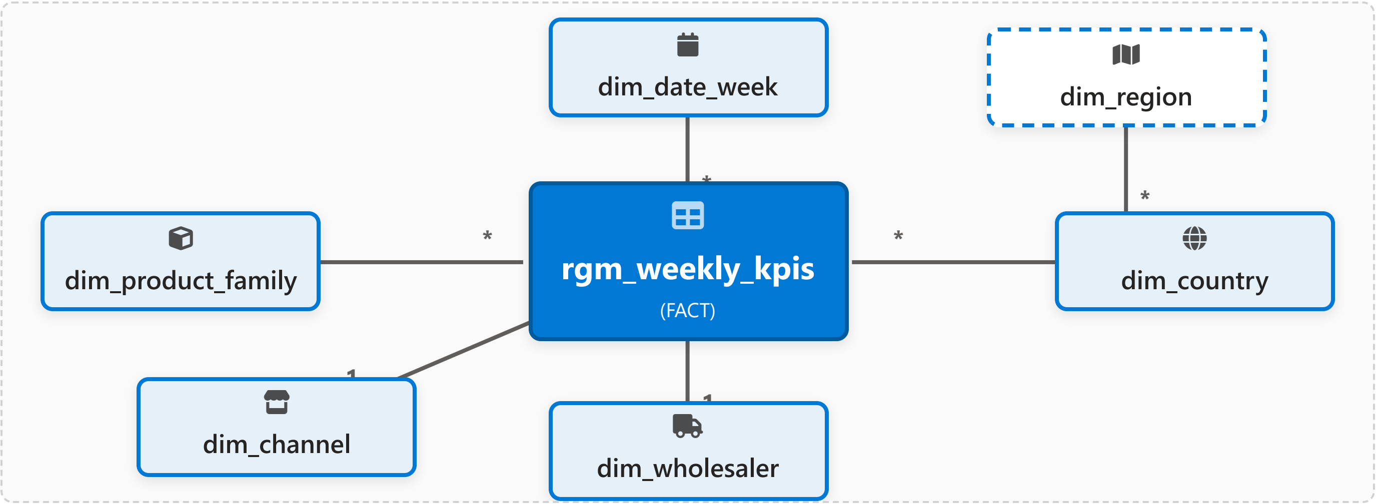
Filter Flow

Single-direction filter flow to ensure predictable propagation.

Business Keys

Business keys used at the analytics layer for clarity.

Relationship Topology



Excluded Relationships

- No direct relationship: Fact → Region (flows through Country)
- No direct relationship: Fact → Detailed Product (avoids many-to-many)

Wave 1 — Core Business Measures

Purpose & Implication

Foundation for all diagnostic and trend analyses. Measures respond dynamically to all context filters.

Technical Implementation

All metrics implemented as DAX measures (not calculated columns) ensuring flexibility and reuse.

REVENUE

Gross Revenue

Total invoiced sales before any deductions or trade spend adjustments.

TRADE

Trade Spend

Total investment in trade promotions, discounts, and rebates.

REVENUE

Net Revenue After Trade

Gross Revenue minus Trade Spend. The primary "clean" revenue metric.

VOLUME

Shipments Units

Total quantity of products shipped in standard units.

PRICING

Average List Price

Volume-weighted average of the gross list price per unit.

PRICING

Average Net Price

Volume-weighted average of the net price (after trade) per unit.

EFFICIENCY

Trade Spend %

Trade Spend as a percentage of Gross Revenue. (Spend / Gross).

PRICING

Price Realization

The percentage of list price realized after trade spend. (Net / List).

MARGIN

Contribution Margin Proxy

Estimated margin after standard COGS and trade deductions.

Wave 2 — RGM Diagnostic Measures

Diagnostic Intelligence Library

Revenue per Unit

Average revenue generated per individual unit sold. Key baseline for assessing value realization across portfolio.

Trade Spend per Unit

Average trade investment allocated per unit. Helps identify over-invested SKUs relative to their price point.

Price Leakage (Abs/%)

Quantifies difference between list price and net realized price. Highlights where value is eroding most significantly.

Trade Efficiency Index

Composite score measuring revenue lift relative to trade investment. Higher scores indicate more effective spend.

Units per Trade Dollar

Volume return on investment. How many incremental units are moved for every $1 of trade spend utilized.

Price vs Volume Signal

Categorical flag identifying scenarios where discounting exists but volume response is weak or negative (Red Flag).

Diagnostic Logic & Interpretation

Designed to explain *why* performance changes occur by isolating key drivers.

- → Identify pricing erosion hot-spots
- → Pinpoint inefficient trade spend
- → Detect revenue quality dilution
- → Assess volume elasticity to discounts

Validation Observations

- **Logic Integrity:** Validated strictly at weekly grain. Consistent with dataset design.
- **Data Consistency:** No segments exhibited discounting without volume support in current data (No Red Flags).
- **Behavior:** Absence of anomalies reflects coherent trade/pricing behavior, not modeling errors.

Wave 3 — Weekly Trend & Monitoring

Purpose & Implication

Smoothing weekly volatility to highlight true performance momentum. Enables early detection of pricing deterioration or trade inefficiencies before they become quarterly issues.

Validation Status

No cumulative distortions or filter conflicts observed. Rolling calculations correctly handle fiscal week boundaries and year-end transitions.

ROLLING 4-WEEK (R4W) MEASURES

Gross Revenue (Rolling 4W)
Sum of Gross Revenue for the current week + previous 3 weeks.

Net Revenue After Trade (R4W)
Smoothed view of net sales performance.

Shipments Units (R4W)
Volume trend baseline to assess demand stability.

Trade Spend % (R4W)
Ratio of 4-week Trade Spend to 4-week Gross Revenue.

Price Realization (R4W Context)
Trend of net price realization % over rolling period.

WEEK-OVER-WEEK CHANGE

Gross Revenue WoW % — WoW
(Curr Week - Prev Week) / Prev Week

Net Revenue WoW % — WoW
Focus on net growth vs gross growth

EXECUTIVE SIGNAL

Revenue Momentum Signal

Automated classification of short-term performance trend.

↑ Improving | — Flat | ↓ Declining

Measure Organization & Best Practices

MODELING STANDARD

To ensure optimal performance and flexibility, we strictly prioritize **DAX Measures** over calculated columns for all aggregations and KPIs in the semantic model.

DAX Measures
Dynamic, context-aware calculation on query time.

Calculated Columns
Static, increases model size, no context awareness.

ORGANIZATION STRATEGY & BENEFITS

Centralized Storage

All DAX measures are defined once and stored centrally within the **rgm_weekly_kpis** table, creating a clean semantic home.

Universal Reuse

Measures are designed to be reused across all report pages and visuals, ensuring every chart speaks the same data language.

Consistency & Clarity

Provides a professional semantic design where metrics like "Net Revenue" have a single definition across the entire organization.

Ease of Maintenance

Logic changes are applied in one central place and instantly propagate to all dependent visuals, reducing maintenance overhead.

Project Status & Next Steps

Project Status

ANALYTICS COMPLETE

- Data validated and loaded into Power BI
- Semantic model finalized (Fact/Dim structure)
- Core commercial KPIs implemented (Wave 1)
- Diagnostic intelligence added (Wave 2)
- Weekly trend monitoring enabled (Wave 3)

Model ready for visualization & storytelling

Next Phase

Dashboard Layout

Design grid structure and navigation for report pages.

Visual Hierarchy

Implement charts and visuals adhering to IBCS standards.

Executive Storytelling

Finalize narrative flow for weekly commercial reviews.

The Power BI model is now **analytics-complete** and ready for the UI layer.